

الجمهورية العربية السورية
وزارة التربية

English for Starters

Vocational Stream

Industrial Grade 12

Computer and Electronic Technology

Students' Book



٢٠١٨ - ٢٠١٩ م
المؤسسة العامة للطباعة



حقوق التوزيع في الجمهورية العربية السورية
محفوظة للمؤسسة العامة للطباعة

تدقيق ومواءمة كتب اللغة الإنكليزية
لينا الذياب
رشا بايزيد
نبال حنا
سرور شلش
باسل صادق
شهرزاد الفراتي
علا داوود آغا
سمر الشيشكلي



322 Old Brompton Road,
London SW5 9JH,
England

Maktabat El Nashr El Tarbawi El Sourì
(Syrian Educational Publishers)

Omar El Mukhtar 2nd Str., Bldg. 6
El Mazraa, Damascus-Syria
Phone: (011) 44676789
Fax: (011) 44676788
e-mail: info@syrianep.com
www.syrianep.com

© York Press 2018

*All rights reserved; no part of this publication
may be reproduced, stored in a retrieval system,
or transmitted in any form or by any means, electronic,
mechanical, photocopying, recording, or otherwise,
without the prior written permission of the Publishers.*

Contents

Unit	Title	Page
1	Optical Storage	6
2	Digital Cameras	10
3	Power Supply	14
Review	Units 1–3	18
4	Sensors	20
Culture and Values 1	Global Warming	24
5	Maintenance of Home Appliances	25
6	Installation of Lifts and Escalators	29
Review	Units 4–6	33
7	Inside the System	35
8	Robots	39
Culture and Values 2	Women in Business	43
9	X-rays	44
Review	Units 7–9	48
10	Automatic Control	50
11	Electrical Power Station	54
12	Latest Discoveries	58
Culture and Values 3	Desertification	62
Review	Units 10–12	63

Scope and Sequence

UNIT	LANGUAGE	NEW SKILLS
1 Optical Storage (pp. 6-9)	Language Functions: giving examples; stating advantages; giving instructions Vocabulary: terms related to optical storage Grammar: review of the verb forms	listening for details; practising taxonomic skills; analysing a table
2 Digital Cameras (pp. 10-13)	Language Functions: giving explanations; making descriptions Vocabulary: terms related to digital cameras Grammar: future time clauses	matching words with their definition; labelling a picture
3 Power Supply (pp.14-17)	Language Functions: describing a process; giving opinions; giving explanations Vocabulary: terms related to power supplies Grammar: non-defining relative clauses	analysing a flow chart; ordering information

UNIT	LANGUAGE	NEW SKILLS
4 Sensors (pp. 20-23)	Language Functions: expressing obligations and possibilities; stating advantages Vocabulary: expressions with <i>do</i> and <i>make</i> Grammar: modal verbs and the passive	ordering information; practising dictionary skills; summarising; practising research skills
5 Maintenance of Home Appliances (pp. 25-28)	Language Functions: giving explanations; identifying faults; giving examples; making justifications Vocabulary: home appliances Grammar: reported questions	reordering letters to form words related to home appliances; matching words with their definitions; writing about the importance of computer maintenance
6 Installation of Lifts and Escalators (pp. 29-32)	Language Functions: making justifications; giving examples; distinguishing language functions; giving opinions Vocabulary: parts of a lift Grammar: reporting verbs used in the negative; reporting verbs + infinitive	labelling a figure; matching descriptions with their corresponding picture; specifying the main idea in a text

Scope and Sequence

UNIT	LANGUAGE	NEW SKILLS
7 Inside the System (pp. 35-38)	Language Functions: presenting facts; expressing conditions; expressing preferences; making comparisons; giving opinions; giving explanations Vocabulary: words for computer hardware Grammar: the second conditional	listening to identify topics; matching vocabulary with their meaning; describing items; role playing; matching titles with their description
8 Robots (pp. 39-42)	Language Functions: expressing unreal conditions; making comparisons Vocabulary: terms related to robots Grammar: the third conditional	distinguishing between true and false sentences; matching vocabulary with their meaning; joining sentences; completing sentences; writing a paragraph about life now and in the past
9 X-rays (pp. 44-47)	Language Functions: describing a process; asking to confirm information; stating advantages Vocabulary: parts of a refrigerator Grammar: tag questions	completing a gap-fill exercise; categorising words; identifying the main ideas in a text; rephrasing a sentence; identifying the meaning of new words from context

UNIT	LANGUAGE	NEW SKILLS
10 Automatic Control (pp. 50-53)	Language Functions: describing automatic processes; expressing agreement or disagreement; justifying answers Vocabulary: terms related to automatic control systems Grammar: indefinite pronouns	distinguishing between true and false statements; completing a gap-fill exercise; completing a timeline; completing a multiple-choice task; writing a paragraph
11 Electrical Power Station (pp. 54-57)	Language Functions: justifying; comparing and contrasting; describing the first power station; rephrasing Vocabulary: words related to power stations Grammar: prepositions	discussing the influence of electricity on human life; labelling a picture; practising dictionary skills; identifying antonyms
12 Latest Discoveries (pp. 58-61)	Language Functions: giving opinions; justifying; talking about new and important inventions; giving examples; giving explanations Vocabulary: words for important discoveries Grammar: sentence fragments; run-on sentences	preparing a short presentation; researching and writing a paragraph about an invention; matching an interview with its corresponding description; writing an introduction and a conclusion to a text; listening to complete a text; reading descriptions to identify the invention; matching inventions and their titles and categorising them

1 Optical Storage

A Dialogue

- 1 Read the dialogue then answer the questions.

Lab technician: Hello. Today I am going to show you around the lab and I will teach you about optical storage.

Student: We haven't been in a lab like this before. This is going to be very interesting. Can you tell us what technology uses optical storage?

Lab technician: Yes! One example is the CD. As you know, a CD can store a large amount of digital information. The surface of the CD is covered with billions of tiny bumps that circle into a spiral track. The CD player reads these bumps with a laser beam, and that's how the information is read.

Student: Thank you! That's very helpful.

Lab technician: Okay now, let's go around the lab ...

Questions

- Find words in the text which mean:
 - a ray of light.
 - a curve in the form of a line that turns around a central point.
 - delightful or enjoyable.
- Name another appliance that uses optical storage devices.

B Listening

- 2 Put the following sentences in the order you hear them.

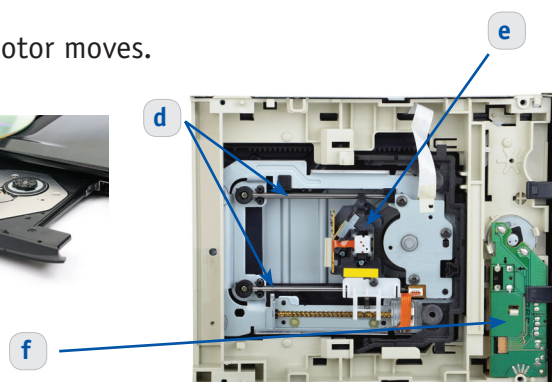
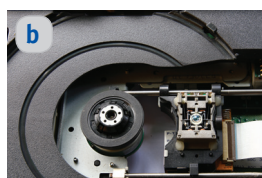
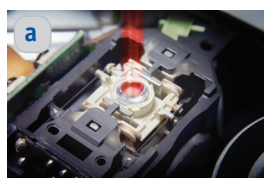
- _____ a Apply white paint to the disc.
- _____ b A laser machine cuts notches into the glass disc.
- _____ c Wash the glass disc in a nickel bath.
- _____ d Cut the glass to the correct size.
- _____ e Apply a layer of polycarbonate and aluminium.
- _____ f Wear special clothing.
- _____ g Coat the glass with photosensitive material.



C Vocabulary

3 The list below shows parts of a CD player. Match the picture with the correct description.

- 1 The **spindle** is a circular part that holds the CD while it is spinning.
- 2 The **tray** is the place where you put the CD.
- 3 The **laser lens** reads the data stored in the bumps on the CD.
- 4 The **circuit board** is a small board on which all the electrical connections are attached.
- 5 The **motor** supplies power for the CD to spin.
- 6 The **rail** is a set of two metal lines on which the motor moves.



D Properties and uses

4 The following table shows some types of disc. Read it and circle the correct answer below.

Type of disc	Features	Storage capacity	Use
CD-ROM	- shiny surface on one side - various colours and designs on label - manufactured with data for people to read only	up to 700Mb	storing music/small amounts of data
DVD-ROM	- shiny surface on one side - various colours and designs on label - manufactured with data for people to read only	up to 4.7Gb	storing films/videos
CD-R	- shiny surface on one side - manufactured blank for people to record data one time only	up to 700Mb	recording music/small amounts of data
DVD-R	- shiny surface on one side - manufactured blank for people to record data one time only	up to 4.7Gb	recording films/videos
CD-RW	- shiny surface on one side - manufactured blank for people to record data many times	up to 700Mb	recording music/small amounts of data
DVD-RAM	- shiny surface on one side - manufactured blank for people to record data many times - lasts a long time (around 30 years)	up to 9.4Gb	recording large amounts of data

1 All types of disc have _____.

- a a shiny surface b a large storage capacity

2 A DVD-ROM can store up to _____ of data.

- a 4.7Gb b 4.7Mb

3 A CD-R and a DVD-R can be used for _____.

- a reading only b recording data several times

4 A DVD-RAM is useful for _____ amounts of data.

- a recording large b recording small

5 All types of CD can _____ 700Mb.

- a record up to b contain up to

c the capacity to record data

c 9.4Gb

c recording data only once

c reading large

c store at least

E Grammar

Review of the Verb Forms	
Simple Present It expresses an unchanging, repeated or recurring action or situation that is happening now. It can also represent a widespread truth. Examples <i>He works as a computer engineer.</i> (unchanging state) <i>Every year, the computer engineer makes a new software.</i> (recurring action) <i>Computer engineers are well known for being very smart people.</i> (widespread truth)	
Simple Past It expresses an action or situation that was started and finished in the past. Most past tense verbs end in <i>-ed</i>. The irregular verbs have special past tense forms which must be memorised. Examples <i>The original format of the compact disc, or CD, stored and played only sound recordings.</i> (regular form) <i>Back in 1982, a CD could store more data than a personal computer hard drive.</i> (irregular form)	
Simple Future It expresses an action or situation that will occur in the future. This tense is formed by using <i>will</i> or <i>shall</i> with the base form of the main verb or by using <i>am</i>, <i>is</i> or <i>are</i> + <i>going to</i> with the base form of the main verb. Examples <i>The science exhibition will finish in a few days.</i> (regular form) <i>The scientist is going to display his inventions in the exhibition.</i> (<i>am</i>, <i>is</i>, or <i>are</i> with <i>going to</i>) <i>The new exhibition starts tomorrow.</i> (the present tense of the main verb with an adverb or adverbial phrase to express future, e.g. tomorrow)	
Present Continuous It describes an ongoing action that is happening at the same time as the statement is being written. This tense is formed by using <i>am</i>, <i>is</i> or <i>are</i> with the main verb form ending in <i>-ing</i>. Example <i>Our teacher is teaching us about the history of the CD player.</i>	
Past Continuous It describes a past action which was happening when another action occurred. This tense is formed by using <i>was</i> or <i>were</i> with the main verb form ending in <i>-ing</i>. Example <i>He was recording his favourite music on a CD when the electricity went off.</i>	
Present Perfect It describes an action that happened at an indefinite time in the past or that began in the past and continues in the present. This tense is formed by using <i>has</i> or <i>have</i> with the past participle of the verb. Most past participles end in <i>-ed</i>. Irregular verbs have special past participles that must be memorised. Examples <i>Researchers have travelled to many countries to collect more data.</i> (indefinite time) <i>Acrylic paints have been available since the 1950s.</i> (continues in the present)	
Past Perfect It describes an action that took place in the past before another past action. This tense is formed by using <i>had</i> with the past participle of the main verb. Example <i>By the time the decorator arrived, the clients had left.</i>	

5 Contrast the meanings of the two sentences.

Example

1 **a** He manufactures two hundred CDs per month.

b He is manufacturing two hundred CDs.

Sentence **a** means that he usually manufactures two hundred CDs per month, whereas in **b** he is manufacturing two hundred CDs now.

2 **a** He has been in the factory for 20 minutes.

b He was in the factory for 20 minutes.

3 **a** When the customer arrived, we discussed the needed quantities.

b When the customer arrived, we had discussed the needed quantities.

4 **a** He promised to give us 2000 SYP, and he has already sent 1000 SYP.

b He promised to give us 2000 SYP, and he had already sent 1000 SYP.

F Reading

6 Read the text then answer the questions.

Using an optical storage device such as a CD or a DVD is a very effective way of storing digital data. CDs and DVDs can last a very long time if you look after them properly. **They** are portable and very easy to store as they are small. They are also cheap to produce. They can be used on many different appliances such as laptops, computers as well as CD players and DVD players.

There are a variety of CDs and DVDs to choose from, depending on what you need them for. You can use a CD-R or a DVD-R to record data one time only, but you cannot rewrite on it. **This** is a very good safety feature that prevents the data from being lost. A CD-RW or a DVD-RAM, however, can be used to record data several times; in other words, **it** can be used again and again. The DVD-RAM is also useful if you want to store a large amount of data as it has a bigger storage capacity than the DVD-R.

A very useful feature of optical storage is that data can be accessed quickly, and it is easy to pinpoint a particular piece of data that had been recorded on the disk.

Computer technicians are usually well-informed in computer hardware and software, including optical storage. They are also knowledgeable about installing and repairing computers, servers, keyboards and printers. They can deal with Internet connection and manage software updates. A good computer technician must have good problem-solving skills and excellent communication skills.



Questions

1 What do the words in bold refer to?

2 According to the text, what are the advantages of optical storage devices? Can you think of another advantage that is not mentioned in the text?

3 What makes a successful computer technician?

4 Do your own research to find out about other optical storage devices.

2 Digital Cameras

A Dialogue

- Read the dialogue and answer the questions below.

Rami: How do digital cameras work?

Fuad: Digital cameras focus light onto a semiconductor to create a digital image.

Rami: I see. Are they similar to conventional cameras?

Fuad: Of course not. Conventional cameras depend entirely on chemical and mechanical processes – you don't even need electricity to operate them. On the other hand, all digital cameras have a built-in computer, and all of them record images electronically.

Rami: So, can we say that digital cameras are better than conventional ones?

Fuad: Not really because film still provides better picture quality, so digital cameras have not completely replaced conventional cameras. But, as digital imaging technology has improved, digital cameras have rapidly become more popular.

Rami: Can I take a picture and email it to a friend if I have a digital camera?

Fuad: Yes, you can since the image in the digital camera is represented in the language that computers recognise – bits and bytes. A digital image is just a long string of 1s and 0s that represent all the tiny coloured dots, or pixels, which collectively make up the image. The digital camera has a series of lenses that focus light to create an image of a scene. But instead of focusing this light onto a piece of film, it is focused onto a semiconductor device that records light electronically. The computer then breaks this electronic information down into digital data.

Rami: Thank you for the information.

Questions

- 1 How are digital cameras different from conventional ones?
- 2 Find words from the dialogue which mean:
 - to give or supply something to someone.
 - a unit for measuring the amount of information a computer can use.
 - extremely small.
 - a picture that you can see through a camera, on a television, or in a mirror.
 - to use something instead of something else.
- 3 What can you do with digital photographs that you cannot do with conventional ones?
- 4 Explain, in your own words, how digital cameras work.



B Listening

2 Listen and answer the questions below.

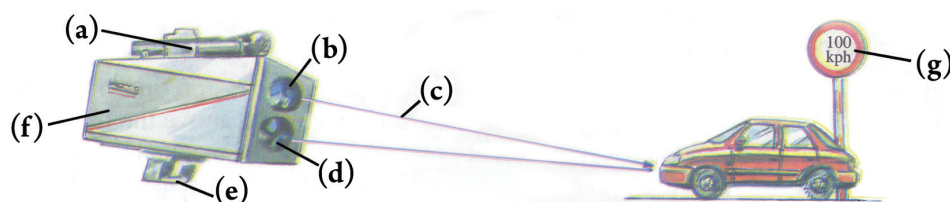
- 1 What is the maximum speed you can drive at on the Syrian highways in order to stick to the speed limit?
- 2 What will the police do to make sure you didn't break the speed limit?
- 3 What will the police do if they catch you speeding?
- 4 How will the speed gun detect if you are going faster than the speed limit?

C Vocabulary

3 Listen again and label the diagram with words and phrases from the list.

telescopic sight	speed limit	speed gun
laser beam	laser transmitter	trigger
		sensor

- a _____
- b laser transmitter
- c _____
- d _____
- e _____
- f _____
- g _____



4 Match the following words with their definitions.

- | | |
|--------------------|--|
| 1 speed limit | a money that must be paid for breaking the law |
| 2 detect | b to work out |
| 3 telescopic sight | c a device that receives a signal of light or motion |
| 4 calculate | d the maximum speed that can be reached |
| 5 fine | e to find out, become aware of something |
| 6 speedometer | f an optical sighting device |
| 7 sensor | g an instrument found in all cars to measure speed |

D Grammar

Future Time Clauses

Time clauses act as adverbs and tell us when something happens. They usually begin with words like: *when, as soon as, before, until, if* and *after*.

Clauses that refer to future time are formed with the verb in the present tense:

Examples

When you drive in Syria, you will have to stick to the speed limit of 100km per hour on highways.

As soon as the police stop you for speeding, they will make you pay a fine.

Before we start working, we will make sure our tools are ready.

After we take the photos, we're going to email them to the participants.

Until all the photos are uploaded, we will have some time to rest.

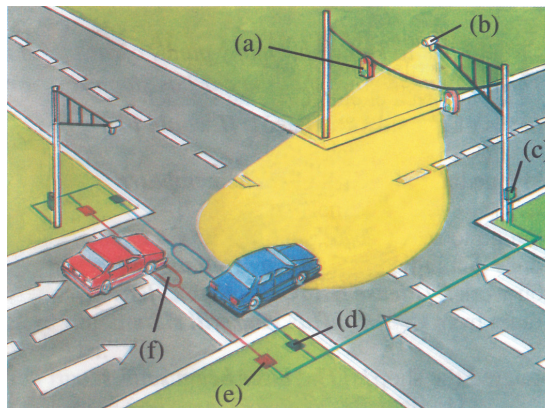
5 Fill in the blanks with the correct form of the verbs.

- 1 I (take) _____ plenty of photos when I (buy) _____ a new camera.
- 2 Until I (fix) _____ my speedometer, I (be) _____ very careful about my speed while driving.
- 3 When he (install) _____ the cameras, he (know) _____ who hasn't been working properly.
- 4 Before we (set) _____ a deadline, we (check) _____ whether we have enough resources.
- 5 As soon as you (break) _____ the speed limit, the police (stop) _____ you and give you a fine.

E Reading

6 Label the diagram with terms from the box. Check your answers after reading the text.

first trigger second trigger sensor loop computer traffic lights red-light camera



7 Read the text and answer the questions.

Red-light Camera System

A red-light camera system detects vehicles crossing a red traffic light. The equipment consists of four cameras, two triggers and a computer. The cameras are positioned on tall poles at the four corners of an intersection and point inwards towards the centre. A trigger is a length of wire laid out in two rectangular coils (called sensor loops) buried under the ground near the stop line. The current through the wire generates a magnetic field. The triggers detect the location of a car and send this information to the computer.

How does the system work? As soon as a car crosses a trigger, it changes the magnetic field around the wire loops. The computer recognises that the magnetic field has changed and knows that a car has passed over the trigger.

There are usually two triggers. The first is at the stop line, and the second is in the intersection. If a car activates both triggers one after the other, the computer will recognise that a car has moved into the intersection. If the car activates only the first trigger, the computer will recognise the fact that it has stopped at the edge of the intersection.

When a car activates both triggers after the light is red, the camera will automatically take a picture that shows the position of the car as it passes the stop line. The camera then takes a second picture. This shows the location of the car in the centre of the intersection. The computer calculates the distance between the two locations. This indicates how fast the car was travelling as it drove through the red light.

Questions

- 1 How many cameras does a red-light camera system consist of?
- 2 How are the cameras positioned?
- 3 Describe what a 'trigger' is.
- 4 How does the computer know that a car has passed over the trigger?
- 5 How can the computer tell if a car has moved into the intersection?
- 6 What is used to indicate how far a car is travelling when it drives through a red light?

3 Power Supply

A Dialogue

- 1 Read the dialogue and answer the questions.

Lab technician: Welcome to my laboratory, **where** I will be teaching you about power supplies. There are two forms of current. AC, or 'alternating current', allows electricity to go in two directions. And DC, or 'direct current', allows electricity to go in only one direction.

Student 1: Where do you find these two forms of current?

Lab technician: The wall outlets in your house usually provide AC power, whereas a DC power supply is a device that can be found in electrical appliances, such as laptops, TVs, refrigerators, and so on. Now, I will explain to you how to convert AC to DC power. The first stage is rectification, when a rectifier converts AC to DC. The rectifier is made from diodes, **which** are semi-conductors that allow electricity to pass one way and not the other.

Student 2: That's it?

Lab technician: Actually, no. You still need to regulate the amount of electricity to the

exact desired level. This is called the smoothing stage. A capacitor, which is like a battery, charges if there is too much electricity coming in and discharges if there isn't enough electricity. Finally, regulation is the last stage, **where** a regulator is used to provide a constantly regular output of power.

Student 3: That's easy. Do you think we can build a DC power supply?

Lab technician: Of course, **that** is what we will be doing next ...

Questions

- 1 What do the words in bold refer to?
- 2 Name the main stages of converting AC to DC power.
- 3 How do you think the capacitor protects home appliances from being damaged?



B Listening

2 Listen and complete the text.

There are many important things to consider for choosing the right (1) _____. First, you need to calculate the energy (2) _____ of each component. The next thing is to figure out the type of (3) _____ you need. Then you have to check the power supply's (4) _____ in converting AC power to (5) _____. High quality is another important factor. It includes the lifespan and the reduction of noise that is generated by the fans for (6) _____ the power supply unit. The difference between quiet and noisy units is the optimal (7) _____ of a fan for a specific (8) _____.

C Vocabulary

3 Complete the text with the correct words from the box below.

input output voltage specification
estimated switching efficient outlet

The switch-mode power supply, which is one of the types of DC power supply, is very (1) _____ at converting electrical power and is used to power PCs, TVs and other appliances. A switch-mode power supply has a (2) power-_____ stage and a control circuit. It is able to supply power from the wall (3) _____ to the appliance and also convert (4) _____ and current. A detailed (5) _____ is needed in order to build a switch-mode power supply. It is important to know the required (6) _____ and (7) _____ ranges before you start building it. You also need to know the (8) _____ total power that is required.



D Grammar

Non-defining Relative Clause

Non-defining relative clauses give extra information about a noun. They are always separated from the rest of the sentence by commas. Compare the following sentences:

Examples

You forgot to add a regulator in order to finish building the power supply. A regulator is an electronic circuit used to provide a stable DC voltage.

You forgot to add a regulator, which is an electronic circuit used to provide a stable DC voltage, in order to finish building the power supply.

Relative Pronouns

	Person	Thing	Place
Subject	who	which	
Object	who / whom	which	where
Possessive	whose	whose	

1 The preposition in non-defining clauses can go at the end of the clause.

Example

The new manager was respected by the people whom he worked with.

This pattern is often used in spoken English, but in written or formal English you can also put the preposition before the pronoun.

Example

The new manager was respected by the people with whom he worked.

2 Non-defining clauses can be introduced by expressions like *example of*, *all of*, *many of*, *both of*, *each of*... + relative pronoun (*whom* for persons and *which* for things).

Examples

There were a lot of people at the meeting, many of whom I had known for years.

He was removing the unsold goods, all of which were spoiled.

3 The relative pronoun at the beginning of a non-defining relative clause can refer to all the information contained in the previous part of the sentence, not just to one word.

Example

A lot of kitchen appliances have a power supply, which I wasn't aware of before. (= The fact that a lot of kitchen appliances have a power supply was new information to me.)

4 Combine the second sentence with the first using a non-defining relative pronoun.

- 1 A switch-mode power supply is a very efficient power supply. It is used to power PCs and TVs.
- 2 A DC power supply has three important main components. They are the rectifier, the capacitor and the regulator.
- 3 The lab technician is showing the students around the lab. The purpose of this is to teach them how to build a DC power supply.
- 4 The power flow in a DC power supply is controlled by a capacitor. A capacitor is a part that keeps the power supply from surging.



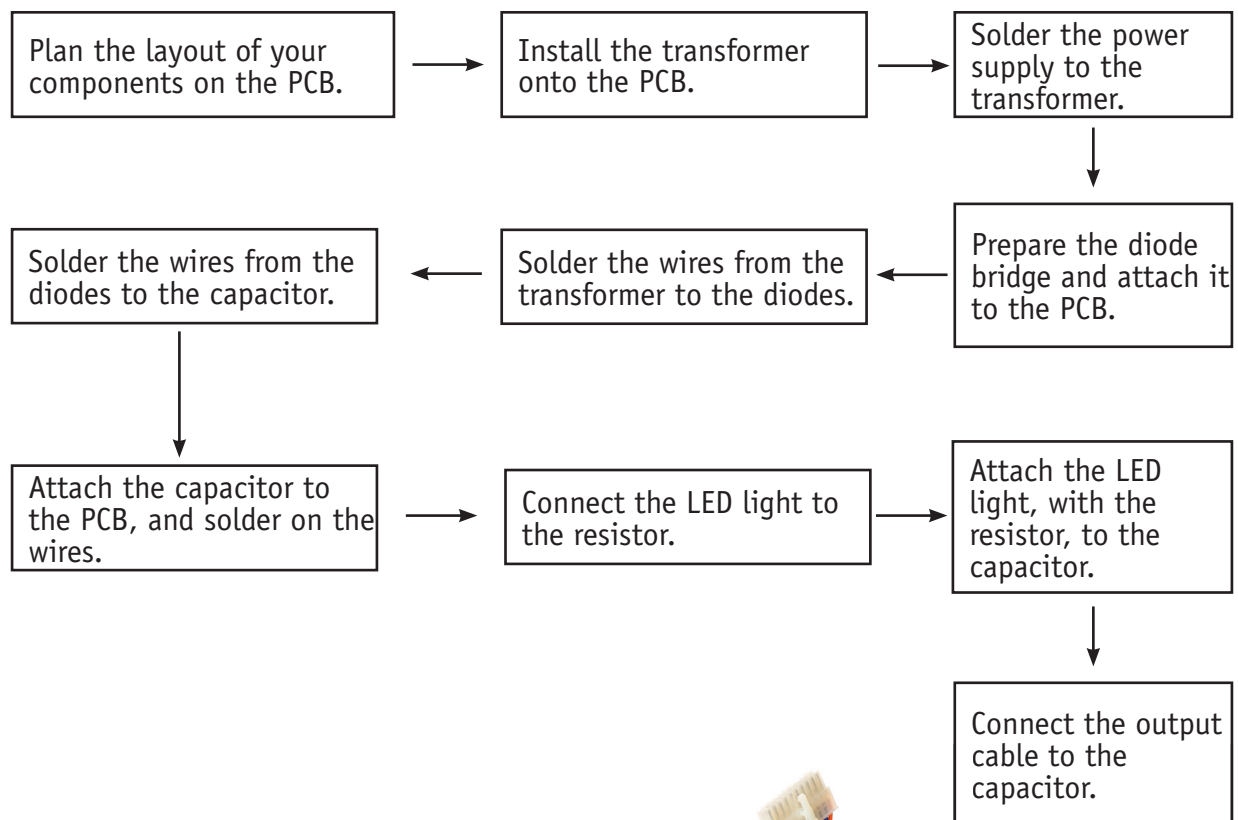
Power Supply

E Reading

5 Read the text and answer the questions.

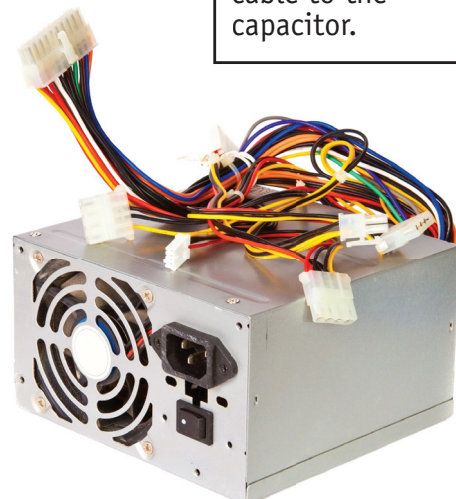
To make an AC power supply, you need

- a Printed Circuit Board (PCB), a **soldering** iron, and copper wires;
- input and output cables;
- a transformer, to reduce the input **voltage** to a lower output voltage;
- a diode bridge, to convert AC to DC (when and if needed);
- a **capacitor**, to smooth the output voltage;
- and a LED light with a **resistor**, to see if it is working properly.



Questions

- 1 Give the text a suitable title.
- 2 What do the words in bold mean?
- 3 Find two words in the text that mean 'to join small pieces together'.
- 4 Use the flow chart to explain to your classmates how an AC power supply is set up.



Review (Units 1-3)

A Match the following words (1–8) with their meanings (a–h).

- | | |
|--------------------|-------------|
| 1 power supply | 5 CD-RW |
| 2 optical storage | 6 speed gun |
| 3 red-light camera | 7 trigger |
| 4 rectifier | 8 CD player |

- a a device used to measure the speed of moving objects such as vehicles
 b a device used to supply electric power
 c a device that turns AC power into DC power
 d a digital camera that detects vehicles that cross a red traffic light
 e the process of storing information digitally
 f a type of disc that can be used many times to record different data
 g a machine that can read optical storage devices
 h a wire that detects the location of a car and sends the information to a computer

B Put the following words under the correct heading.

spindle disc voltage sensor loop traffic lights spiral track
 magnetic field capacitor resistor DVD-R laser transmitter record
 optical pixels rectification current DC trigger

Power supply	Optical storage	Digital cameras

C Fill in the blanks with words from the box.

DVDs drives capacities damaged storage space store

Have optical (1) _____ devices become a thing of the past? People nowadays need to (2) _____ a lot more data than in the previous years. Not only do they want to have their own music library, but they also want to collect their favourite videos, including films, which need a lot of storage (3) _____. This is why almost everyone has been making use of external hard (4) _____. Although they are much more expensive than CDs and (5) _____, external hard drives are available in much larger (6) _____ – up to 1 Tb (1,000 Gb). They are also safer than optical storage devices, which can be easily scratched and (7) _____. Who knows when optical storage devices will become as outdated as the floppy disc?!

D Choose the correct words between brackets.

- 1 The (efficient, efficiency) of some devices is better than others.
- 2 (Digit, Digital) watches show the time in the form of numbers.
- 3 The (electric, electricity) current will be cut off if you don't pay your bill.
- 4 The microwave is much faster than the (convention, conventional) oven.
- 5 It's (important, importance) that doctors explain everything to their patients.

E Rearrange the steps in order.

- 1 The camera takes one photo of the car automatically when it passes the red light and another photo when the car reaches the intersection.
- 2 The camera detects the action of the vehicle crossing the light and moving into the intersection.
- 3 A computer calculates the distance between the two locations.
- 4 A vehicle crosses a red traffic light.
- 5 The magnetic field around the wire loop changes.
- 6 Two triggers are activated.

F Fill in the blanks with the correct form of the verbs in parentheses.

- 1 Every week one of the best students (visit) _____ the laboratory to do an advanced experiment.
- 2 There (be) _____ many types of disc such as the CD-ROM, the DVD-ROM, the CD-R and the DVD-RAM.
- 3 Right now, the teacher (demonstrate) _____ how a digital camera works in front of the students.
- 4 While he (record) _____ music tracks on a CD, an error occurred.
- 5 That man wasn't a novice. He (be) _____ a computer technician for over fifteen years.

G Link the following sentences using future time clauses to form a paragraph about using a digital camera.

- Decide what you want to take a photo of.
- Choose a good angle and focus the camera on the scene or object.
- Take the photo.
- Upload the photo onto your laptop.
- Save the photo in 'your photos' and give it a name.
- Edit the photo for artistic effects.
- Print and frame the photo.
- Happy memories!

H Combine the two sentences using a relative pronoun.

- 1 Efficiency in a power supply is a major concern. It's based on the size, design and material.
- 2 Traffic is very well controlled on every street of this city. Red-light camera systems are installed.
- 3 Dr Jill will teach physics. I met Dr Jill last week.
- 4 Mr Samer explained all the possible ways of storing digital data. He is a competent teacher.

4 Sensors

A Dialogue

1 Read the dialogue then answer the questions.

Salah: Good morning, this is Salah speaking. How can I help you?

Caller: Good morning. I am trying to install parking sensors on my car, and I need some help. Who should I talk to?

Salah: I am just the person you need.

Caller: Great! Usually, how many sensors must be installed?

Salah: Either two or four sensors. **It** depends on the length of your car bumper.

Caller: Oh, okay! Then my car bumper has to be checked first. And can the sensors be placed anywhere on the bumper?

Salah: No, they must be placed on the part of the bumper **that** is flat so that they are not at an angle. Otherwise, they will not work correctly. Also, **they** need to be facing straight so that they can accurately detect objects behind the car.

Caller: That's a little bit delicate. It seems that this kind of job should be done by a specialist. Can I bring you my car tomorrow morning?

Salah: No problem. We'll be waiting for you. Thank you for calling.

Questions

- 1 What do the words in bold refer to?
- 2 Did the caller decide how many sensors should be installed on his car? Why or why not?
- 3 Why is the installation of car sensors delicate?

B Listening

2  Listen and number the following steps in the order you hear them.

- _____ Install the sensors according to the instruction manual. Make sure they are flat and not positioned at an angle. They should not be pointing upwards or downwards.
- _____ Measure the bumper and decide if you want to install two or four sensors. This will depend on the length of your bumper.
- _____ Test the sensors before driving the car to make sure you have installed them correctly.
- _____ Install the control box. This is usually placed in the boot of the car.
- _____ Mark the areas on the bumper where you want to install the sensors.
- _____ Clean the areas around the holes to get rid of old bits of bumper and polish the area to protect the bumper.
- _____ Install the alarm. This should be connected to the control box.
- _____ Drill holes in the bumper in the areas you have marked.

C Vocabulary

- 3 Complete the text with the correct words from the box below.

installed bumper facing sensors
accurate reverse

While driving my car the other day, I had an issue with my parking sensors. I was trying to park in (1) _____, but I couldn't see behind me; so I used my sensors, which allow me to reverse-park while (2) _____ forwards. The sensors were (3) _____ at the back of my car by a mechanic just last week. As I was reversing, I suddenly heard a loud 'thump' and my car stopped. I was shocked because sensors are usually very (4) _____, but this time they hadn't worked properly and I had reversed into the car behind me. When I got out of my car and walked to the back, I found that my (5) _____ had fallen off along with the (6) _____ on it.

- 4 Look up the meaning of the following words using a dictionary.

- 1 sensor
- 2 install
- 3 detect
- 4 reverse
- 5 bumper



- 5 Study the table, then complete the sentences using the verbs in the right form.

do	make
a sport	arrangements
some work	an announcement
some shopping	a decision
well / badly	an offer
your best	a payment
your hair	a phone call
business	a profit
a job	a suggestion
nothing (i.e. relax)	a mistake
some photocopying	a noise

- 1 It has been a pleasure to _____ business with you.
- 2 No arrangements _____ for the meeting, so there weren't enough chairs.
- 3 After the factory visit yesterday, we _____ some shopping in the evening.
- 4 When the announcement _____, the passengers started to board the ship.
- 5 Have you _____ your work yet?
- 6 At yesterday's meeting, people _____ some suggestions about the conference.

D Grammar

Modal Verbs and the Passive		
Subject	Modal verb	Passive infinitive (be + past participle)
It The food	can could has to had to is going to must should will	be imported be frozen be eaten

6 Put these sentences in the passive form.

Example

1 *You can send this letter.*
This letter can be sent.

- 2 They should wear safety hats in the garage.
- 3 We have to pay this invoice before the end of the month.
- 4 We could see the damaged cars.
- 5 We must repair the sensor at once.
- 6 We had to throw away the damaged part of the sensor.
- 7 They will repair the car sensors within a week.
- 8 We're going to send those four containers by rail.



7 Identify whether each sentence is in the Active or Passive form.

Example

1 *A car needs a lot of maintenance. (Active)*

- 2 The sensors can be installed by a mechanic at a garage.
- 3 You must always check your sensors for any errors.
- 4 Install two or four sensors depending on the length of the bumper.
- 5 The car must be repaired by you when trouble appears.



E Reading

- 8** Read the text then complete the table by ticking the corresponding type of sensor for each characteristic.

Sensors can be found on many things. They have the amazing ability to detect even the slightest movements. You can find them in cars to assist with parking, at home in the burglar alarm system, and in many automatic devices. They can be used to keep lift doors from closing too soon and to switch on the light when you enter a room.

Touch sensors are able to detect physical contact with an object. They are responsive to different kinds of touch like tapping and swiping. They are mostly used in devices, such as smartphones, where **they** can detect the pressure of the finger against the screen. Each touch is recorded by the sensor and sent to software to interpret it.

Humidity sensors can detect the presence of moisture in the air and indicate its level. Humidity can have important effects on humans and manufacturing. This is why it has to be controlled. For example, humidity can have an effect on the health and safety of people.

Ultrasonic sensors, a more advanced type of sensor, use sound waves and echolocation. These sensors produce sound waves that bounce back to them when **they** are in contact with something. This triggers the alarm.

Temperature sensors can measure temperature through an electrical signal. They are the most common type of measurement. We can find them in home appliances such as heating systems and refrigerators.

As sensors are part of our everyday lives and are found in so many things we use, it is important to know how **they** work.

touch sensor: Column A	humidity sensor: Column B			
ultrasonic sensor: Column C	temperature sensor: Column D			
Characteristics	A	B	C	D
a detects the presence of moisture				
b measures temperature				
c detects physical contact				
d uses sound waves and echolocation				
e responds to touch				
f can be found in heating and cooling systems				
g can be found in devices like smartphones				

- 9** Read the text again and answer the following questions.

Questions

- Summarise the text in a flow chart.
- What do the words in bold refer to?
- Work in pairs. Search the Internet for information about how sensors are used in hospitals. Discuss your findings with the whole class.

F Writing

- 10** Write a short paragraph discussing some of the benefits of a car sensor. You may look back at previous lessons to help you find ideas.

Culture and Values 1

Global Warming

Climate change is not a new phenomenon in the history of Earth. However, since the beginning of the Industrial Revolution, Earth has started to experience unprecedentedly rapid climate changes due to what we now call global warming.

Global warming is partly caused by harmful gases that are released from burning fossil fuels. Such gases will influence the Earth's climate even years from now, since it has not adapted yet to such a rapid environmental change. The percentage of greenhouse gases is directly proportional to the percentage of fossil fuels that are burned by humans. However, climate change is not only caused by humans; other factors that existed even prior to the Industrial Revolution continue to influence the Earth's temperature.

Increasing temperatures are by no means the only effect of global warming. Global warming also influences annual rainfall, increases the erosion of land that is near the coast, extends the growing season and results in the melting of glaciers. Finally, it can cause changes in the frequency of infectious diseases in particular environments.

Questions

- 1 Rephrase the definition of global warming given by the text and list, in your own words, its effects on Earth.
- 2 What is the meaning of the following words: phenomemon, unprecedentedly, rapid, glaciers?
- 3 Why can't Earth adapt to the changes in climate?
- 4 In your opinion, are humans to be blamed for global warming? Explain using the evidence given in the text.
- 5 Find information about two of the latest solutions for global warming on the Internet (for example: more white clouds needed in the sky and synthetic trees that absorb a bigger quantity of carbon dioxide) and put them with photos you can find in a brochure.

How to prepare a brochure:

- Open a Word document and click «New» in the toolbar.
- Go to «File» and «Page Setup». Select «Margins» and set the margins at half-an-inch for all sides.
- Select «Paper Size». Go to «Orientation» and select «Landscape».
- Click «Format» and then «Columns». Go under «Presets» and select «Three». You will see a section labelled «Spacing». Enter one inch for the spacing and click «OK». You will now have the layout for the brochure.
- Add text and images to the brochure. Type inside the brochure to add text. To insert an image, click «Insert» and then «Picture». Select «From File» and you will be able to use an image from your computer. To add a background, go to «Format» and then «Background». You will be able to select a colour for the background.

5 Maintenance of Home Appliances

A Dialogue

- 1 Read the dialogue then answer the questions.

Laila is calling the "Home Appliances Helpline" to ask the operator for assistance.

Operator: Home Appliances Helpline. How can I help you?

Laila: Hello. I'm calling about my washing machine. It seems to be filling up with water, but it's not running properly.

Operator: Have you made sure that the door is closed tightly after you put your load in?

Laila: Yes, I have. I've even checked that it's switched on properly, but it still doesn't work.

Operator: Then maybe you're overloading the machine.

Laila: Oh dear! What should I do then?

Operator: If you put in lighter loads that should help get your washing machine started. And try to wait around 15 minutes between loads so that the motor can rest.

Questions

- 1 What appliance does Laila have a problem with?
- 2 What is going wrong with the appliance?
- 3 What are the two suggested causes of the problem with the appliance?

- 4 What did the operator tell Laila to do to fix her appliance and allow it to run smoothly?

B Listening

- 2  Listen to some tips on how to keep a computer running smoothly. Number the sentences below so that they are in the correct order. Add the missing tip.

- _____ a Always back up or save any data you don't want to lose, preferably on more than one hard disk.
- _____ b At least once a month, scan your hard disk and defragment it.
- _____ c Make sure you have a UPS (uninterruptible power supply) for your computer.
- _____ d Try not to have too many programs loading while your computer is starting.
- _____ e Do not turn off your computer before it has shut down.
- _____ f Never unplug devices attached to the computer, known as peripherals, while it is still on.
- _____ g _____

C Vocabulary

- 3 Rearrange the letters in column A to form the name of the home appliance that best matches the description in column B.

Column A	Column B
1 ira oicdionnting	a machine which keeps the air in a closed area cool and dry
2 hwiagsn emichna	a machine used to wash clothes
3 ryred	a machine for drying things, usually clothes or hair
4 evon	a device used for cooking food with electricity or gas
5 fegirorart	a large container which is cooled by electricity to preserve food
6 cravimoew	a relatively small appliance which cooks food using electromagnetic radiation
7 rendelb	an electrical kitchen appliance that is used to mix liquids and soft foods



D Grammar

Reporting yes / no Questions
We report yes / no questions with 'if' or 'whether'. DIRECT SPEECH: "Do you need help fixing the oven?" REPORTED SPEECH: I asked him if he needed help fixing the oven. NOTE We do not use the auxiliary verb 'do', except in negative questions.
DIRECT SPEECH: "Who doesn't need new air conditioning?" REPORTED SPEECH: She asked me who didn't need new air conditioning.
Reporting wh- Questions
When we report questions with 'who', 'what' or 'which' + 'to be' + complement, the verb 'to be' can come before or after the complement. DIRECT SPEECH: "Who is the new handyman?" REPORTED SPEECH: She asked me who the new handyman was.

Maintenance of Home Appliances

4 Finish the sentences using reported speech. You may need to change the tense.

- 1 **Ahmed:** "Are the maintenance employees working efficiently?"
Ahmed asked me _____.
- 2 **Jason:** "Who sent the laptop for maintenance?"
Jason wanted to know _____.
- 3 **Robert:** "Is Tim trying to fix the air conditioning?"
Robert asked me _____.
- 4 **Hind:** "Will they provide us with maintenance services tomorrow?"
Hind asked me _____.
- 5 **Rima:** "Where do you send your home appliances to be repaired?"
Rima wanted to know _____.
- 6 **Nancy:** "Why didn't Nick send the broken TV back to the store?"
Nancy wanted to know _____.
- 7 **Huda:** "Where are the best maintenance stores in town?"
Huda asked me _____.
- 8 **Linda:** "Did your mum fly to London two weeks ago?"
Linda asked me _____.
- 9 **A man:** "When does the handyman arrive?"
A man asked me _____.

E Reading

5 Read the text and answer the questions.

A Repair Technician's Job

Home appliance repair technicians have very demanding jobs. They are expected to install and repair home appliances, and are usually required to drive to the location of the equipment. When installing a major appliance like a washing machine or a refrigerator, these technicians may be asked to show the customer how to use it or answer any questions they may have about maintaining the appliance.

The job of a home appliance technician does not only have to do with setting up the equipment, but also repairing it. They are called by customers who have problems with their appliances, and are asked to visually inspect the device and make any necessary repairs.

As a result, this job includes a lot of driving around from one location to another. Sometimes visiting customers and attending to their needs may come up, and the technician is required to deal with these problems as soon as possible. Their job also requires them to constantly update their skills and their knowledge about new models and changes in appliances. This is important for their careers so that they are able to repair the latest appliances. But even though home appliance technicians have tough jobs, they still enjoy helping others.

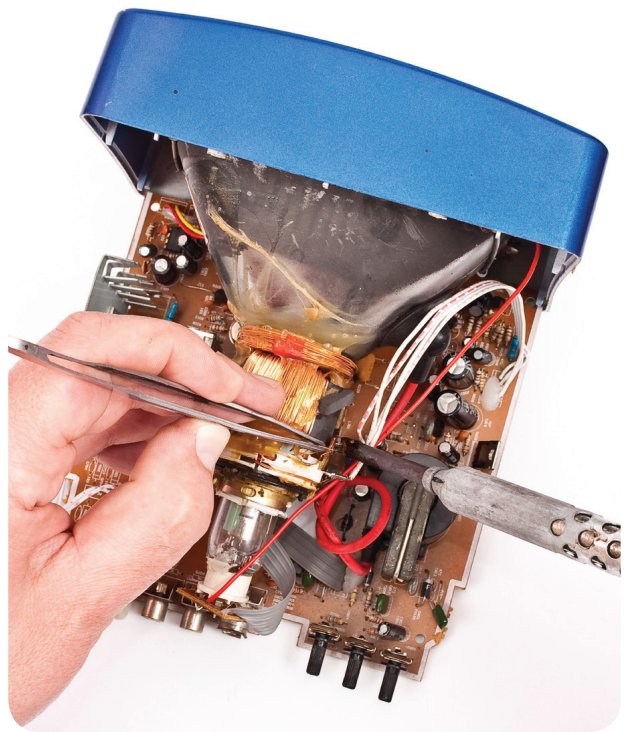
Questions

- 1 Is the job of home appliance repair technician easy? Give examples from the text to support your answer.
- 2 Why should technicians always update their skills and knowledge?
- 3 Define the following words:
a repair c maintaining
b installing d inspect



F Writing

- 6** In this lesson you have learned about the importance of maintaining computers and home appliances. You have also seen what the job of a repair technician is. Would you ever consider working as a home appliance repair technician? Write a paragraph discussing whether you would like this job and why or why not. Refer back to the lesson for help and ideas if necessary.



6 Installation of Lifts and Escalators

A Dialogue

- 1 Read the dialogue and then answer the questions.

Hady: So, Saber, what is it that you do?

Saber: Install and repair lifts.

Hady: What exactly is that?

Saber: I assemble, install and repair lifts.

Hady: Wow! Has your job been affected by the global economic crisis?

Saber: My work isn't affected by changes like that because it relies on maintenance and repair.

Hady: That's great! Maybe I should consider going into the lift installation business.



Questions

- 1 What does Saber do for a living?
- 2 Is Saber's job stable or is it affected by the economy? Why or why not?
- 3 Correct the following false sentences.
 - a Hady has the same job as Saber.
 - b A lift technician's job involves cleaning lifts.

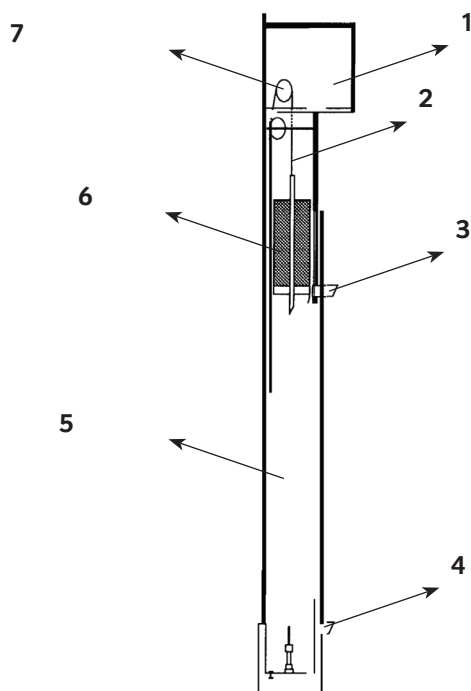
B Listening

- 2 Listen and complete the table.

Hydraulic	Both	Traction
connected to a jack	have a cabin or (3) _____	connected to a (4) _____
operated by a (1) _____		operated by an electric motor
located at the bottom of the shaft		placed (5) _____
found mostly in (2) _____		is the most commonly used form

C Vocabulary

- 3 Label the figure of a traction lift with the terms defined below.



- a Top floor: the last level of a building where access above this floor stops for ordinary passengers
- b Lift shaft: the long empty space through which the cab moves up and down
- c Electric hoist motor: the motor that powers the hoist cable, using electricity to pull it up and lower it down
- d Lift pit: the lowest part of the lift system, located below the shaft
- e Machine room: the room located at the top of the lift shaft that contains the operating motor
- f Hoist cable: the wire that lifts and lowers the lift cabin
- g Cab: the cage or car which carries passengers up and down the lift system

D Grammar

Reporting Verbs Used in the Negative

With a small number of reporting verbs, you usually make the reporting verb, not the reported clause, negative.

believe expect feel imagine propose suppose think

Example

"I expect the company's retail food revenue will not rise much."

He / She does not expect that the company's retail food revenue will rise much.

Installation of Lifts and Escalators

Reporting Verbs + Infinitive			
Many functions are reported with the following pattern:			
Function	Reporting Verb	Object	Infinitive
advising requesting instructing warning encouraging reminding	She advised He asked You told They warned We encouraged I reminded	me her us him them you	to buy insurance. to work late. to send the money. not to drive fast. to try again. to bring the key.
In the examples above, the reporting verb introduces an action.			
If the verb introduces information, a different pattern is used:			
1 The doctor told him to use escalators instead of taking the stairs. (action)			
2 The doctor told him that escalators are not tiring like the stairs. (information)			

- 4 Rewrite these statements as negative reported speech using the verbs in brackets.

Example

- 1 Tom Sharp doesn't understand how to install lifts. (feel)
I don't feel that Tom Sharp understands how to install lifts.
- 2 Tom has not had much experience in installing escalators. (believe)
- 3 We should not look for a new employee who has more experience in repairing lifts. (propose)
- 4 We could always ask for a different manager. (believe)
- 5 They would not agree to this in the middle of a critical state of sales. (expect)
- 6 They'll understand our position. (think)



- 5 Identify the function. Then report the speech using the correct verb.

Example

- 1 Ali: "Would you please phone the manager, Huda?"
Requesting. Ali asked Huda to phone the manager.
- 2 Mike: "Laila, don't leave the escalators working after 6 p.m. please."
- 3 Susie: "Why don't you take the lift instead of the stairs, Alice? I think you'll save some time and energy."
- 4 Ali: "Jane, remember to press the alarm button if the lift suddenly stops."
- 5 Huda: "Don't go down an ascending escalator, Ali!"
- 6 Khaled: "If I were you, I would install more escalators in the shopping centre."
- 7 Samer: "Please don't go into an overcrowded lift, Liz."

E Reading

- 6 Read the text and answer the questions.

Lift installation requires a lot of knowledge in many technical fields. Since a lift is a form of transport that is powered by an electric motor, it is important to know about the motor as well as the rules of mechanics. Some lifts need particular fluids to operate, while others use a pulley system. Because of this, an installation technician should be able to understand both the mechanical and the electrical parts of the system. Lift mechanics are skilled enough to adjust or fine-tune the equipment. They need to know not only about electronics, but also about computers, which they use in their work. Most installers are educated through apprenticeship programmes that allow them to train under professionals. All in all, we can say that lift installers need a lot of knowledge and skills to be able to do their job properly.

Questions

- 1 What should lift mechanics be educated in?
- 2 What is the main idea of the passage?
- 3 Give the passage a suitable title.
- 4 Find words in the text that can replace the following words:
 - a needs (v)
 - b liquid
 - c training
 - d correctly

F Writing

- 7 Installing a lift isn't an easy job. As we have seen, the specialised technician must have a lot of knowledge and experience. In your opinion, what subjects should a lift mechanic have been good at in school? Write a paragraph discussing your answer. You may use the word bank below to help you as you write.

Maths	apprentices	motors	educate
Physics	electronics	traction	training
Chemistry	mechanics	hydraulic	construction trade
Biology	skilled	operate	repair
English	field	adjust	lifts

Review (Units 4-6)

A Match the following words to their meanings.

- | | |
|-------------------|---|
| 1 sensor | a to cause something to happen |
| 2 ultrasonic | b a device which detects heat, light, motion or pressure |
| 3 echolocation | c to make small adjustments to come to the desired outcome |
| 4 trigger | d the act of learning a certain skill or trade from an experienced employer |
| 5 assist | e to help |
| 6 slightest | f demanding a lot of effort or skill |
| 7 fine-tune | g describing sound waves above the level of human hearing |
| 8 demanding | h to put in a position ready for use |
| 9 install | i smallest |
| 10 apprenticeship | j the position of an object by reflected sound |

B Construct a sentence with each word.

- 1 measure _____
- 2 bumper _____
- 3 assemble _____
- 4 repair _____
- 5 rely _____

C Put the following words under the correct heading.

washing machine shaft dryer install parking reverse
hoist repair maintenance microwave alarm system pulley

Home appliances	Sensors	Lifts and escalators	Words in common

D Complete these sentences following the given example.

The bumper should be measured before installing the sensors.

- 1 The car must be checked by a mechanic when _____
- 2 The bumper ought to be cleaned when _____
- 3 The sensors should be tested before _____
- 4 You can install sensors that can switch on the light when _____

E Identify four types of sensors. What are they used for?

- 1 _____
- 2 _____
- 3 _____
- 4 _____

F Decide if the sentences are true (T) or false (F). Correct the false sentences.

- 1 A home appliance repair technicians' job is very easy.
- 2 The technicians must update their skills and knowledge about new models and changes in appliances.
- 3 A repair technician's job is repairing home appliances.
- 4 Repairmen drive from one location to another according to customer demands.

G Identify the seven parts of a traction lift.

H Complete the following clauses.

- 1 Lift installation requires _____
- 2 A lift is a form of _____
- 3 A lift installer should know about _____
- 4 Most installers are educated through _____

I Fill in the blanks with *make* or *do* using the verbs in the right form.

- 1 I have _____ arrangements to have your car serviced next week.
- 2 The home appliance repair technician _____ a mistake while installing the washing machine.
- 3 Mechanics must _____ sports to remain fit to be able to move swiftly while working.
- 4 _____ business with competent handymen is always a pleasure.
- 5 Your hair looks marvellous! Where have you had it _____ ?

J Put these sentences in the passive form.

- 1 Salah will make an appointment to have parking sensors installed.
- 2 You can't measure the temperature of a faraway object using a contact thermometer.
- 3 Professional technicians must supervise the installation of a home alarm system.
- 4 Touch sensors can detect physical contact with an object.

K Identify whether each sentence is in the *active* or *passive* form.

- 1 You can install your washing machine if you read the instructions on the manual carefully.
- 2 A lift mechanic ought to be educated.
- 3 A lot of experience is needed to install escalators.
- 4 Repair technicians should update their knowledge and skills.
- 5 Technicians who install lifts should understand the mechanical and electrical parts of the system.

L Change these sentences to reported speech and say whether the reporting verb introduces an action or information.

- 1 The doctor said to the patient: "Eat healthy food and exercise daily."
- 2 Our teacher reminded us: "Study well for the test."
- 3 The teacher said to the students: "You have to go over the information several times."
- 4 Nora said to the teacher: "I would like to use the dictionary during the test."
- 5 The repairman said to the lady: "Don't touch anything because you may damage the microwave."

M What would you like to be in the future, a repair technician, an engineer working with sensors or a lift technician? Write five lines.

7 Inside the System

A Dialogue

- 1 Read the dialogue and choose the correct answer.

Shop assistant: Hello, sir. May I help you?

Salah: Hello. I'd like to buy a new computer for my son, but I am not familiar with computer hardware.

Shop assistant: I'll be glad to help you, sir. What does your son need the computer for?

Salah: I'm sure he will use it to play games, but he mainly needs it for homework and school projects. And he definitely needs to connect it to the Internet.

Shop assistant: Okay, would you like to get him a laptop or a desktop computer?

Salah: I think he would prefer a laptop. It is more practical, and he might need to take it to school one day to present his projects!

Shop assistant: Okay, then. We have a good selection of laptops right here. For the work he will be doing, I think a 500Gb hard drive would be enough. As for the RAM, I used to work on one with only 2Gb and it was very slow. If I were you, I would get one with 4Gb.

Salah: I think you're right.

Shop assistant: Which brand do you prefer?

Salah: I don't mind any, as long as it's affordable. This one has a reasonable price.

Shop assistant: And it's doing well in the market! I would recommend it.

Salah: Okay, then. I'll buy it!



Questions

- 1 Salah needs help in choosing a computer because
 - a he doesn't know how to use a computer.
 - b he has never bought a computer.
 - c he doesn't know much about computer hardware.
 - d he has a lot of options in mind.
- 2 Salah says a laptop is a better choice for his son because
 - a he might need to take it to school.
 - b he will play games on it.
 - c he needs to connect it to the Internet.
 - d he needs it for homework.
- 3 Salah chooses a laptop with 4Gb RAM because
 - a it costs less.
 - b one with only 2Gb will be too slow.
 - c the shop doesn't have one with 2Gb.
 - d the laptop won't work with 2Gb RAM.
- 4 Salah decides to buy
 - a the cheapest brand of laptop.
 - b a laptop that is good but not very expensive.
 - c the best brand of laptop.
 - d a laptop that is good and expensive.

- 2 What are the features to look for when buying a computer? In pairs, make a list of the features that you know about.

B Listening

- 3 Listen to the dialogue between Mike and a computer technician. Which four of these topics are discussed? Number them in order from 1–4.

A Hard drive _____ B USB _____ C HDMI cable _____
D Computer manual _____ E Separate screen _____ F RAM _____

- 4 Listen again. Write one fact for each of the four topics.

C Vocabulary

- 5 Read the brochure about laptops. Then match the underlined words with their definitions (1–8).

Personal computers, or PCs, can be desktop computers or laptops. Desktop computers are stationary, whereas laptops are portable, and so they are lighter and smaller than desktop computers. However, the essential internal parts of both types are the same. The CPU is the brain of the computer. It carries out the instructions of a computer program. The faster the CPU, the quicker the computer works. The motherboard is the circuit board inside the computer where the CPU is held. The motherboard allocates power and allows communication between the CPU, the RAM and the other parts of the computer hardware. To protect the motherboard and the CPU, solid casing is required, especially for laptops, which are lightweight and can easily crack.

- 1 built to weigh less than average
- 2 important, necessary
- 3 not moving, or not intended to be moved
- 4 to distribute for a certain purpose
- 5 the part that controls the computer operations (Central Processing Unit)
- 6 a cover that protects something
- 7 the hardware that controls the memory of the computer (Random Access Memory)
- 8 the circuit board inside the computer

D Grammar

The Second Conditional	
We use the second conditional if the condition is impossible now or in the future:	
Condition	Result
<i>If I were you,</i>	<i>I would work harder.</i>
If the speaker thinks the condition is realistic or likely, the first conditional is used.	
Example <i>If you reduce your tariffs, we will reduce ours.</i>	

- 6 Complete the sentences using the verbs in brackets.

- 1 If I (know) _____ how to make this product more useful, I would tell you.
- 2 I would buy a faster laptop if I (be) _____ you.
- 3 If we (not take) _____ the customers' interests into consideration, our success in computer production would not be the same.
- 4 If it (be) _____ possible, I would provide you with the final sample of the new computer screen tomorrow.
- 5 It would cost less if we (cooperate) _____ with other computer technicians.
- 6 If we (have) _____ a million Syrian pounds, we would establish a new computer factory.
- 7 If this new type of hardware (be) _____ slow, nobody would buy the product.

E Preferences and comparisons

- 7 Are these shoppers talking about general things (G) or specific things (S)? Mark the phrases by 'G' or 'S'.

Examples

- 1 I like those ones more. S
 2 Red is not my favourite colour. G
 3 I like it more than the green one.
 4 I don't like desktop computers; I prefer laptops.
 5 I don't like it as much as the green one.
 6 I prefer light colours to dark ones.
 7 I don't like it.
 8 What about this one / these ones?

- 8 Work in pairs. Imagine that you are shopping with a friend. Express your preferences over certain objects and make comparisons.

Example

- A What about those sunglasses?
 B No, I don't like them. I prefer lighter frames.

F Buying a product

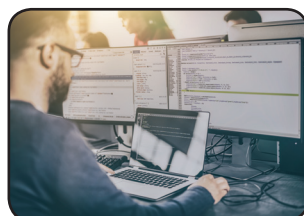
- 9 Look at the information in the table below. Which specifications should each of the following people choose for a new laptop? In pairs, do a roleplay making comparisons and expressing preferences.

- **Student A:** a student doing school projects
- **Student B:** a web designer
- **Student A:** a photographer
- **Student B:** a teenager playing computer games

Specifications	Options	Who chooses these options?
CPU	2 GHz	people who can deal with a fairly well-performing laptop
	4 GHz	people who need a very good performance for their kind of work
RAM	4 Gb	people who perform simple tasks like surfing the Net, typing documents, etc.
	8 Gb	people who perform more complex tasks like working on photos, videos, professional software, etc.
Hard drive / SSD	500 Gb hard drive	people who want a fairly fast laptop
	250 Gb SSD	people who want to store a big amount of files; professional people whose work requires a lot of storage space

Example

- A I think a 2 GHz CPU is enough for doing my school projects.
 B For me, as a web designer, I prefer a 4 GHz CPU. I need a better performance.



G Reading

10 Match the titles with the descriptions. Then answer the questions.

a Microphone Socket

b Serial Port

c HDMI Port

d Audio Socket

e Power Cable Port

f Ethernet

g USB Port

1 _____

This port is the one you use if you want to connect your computer to a screen so that large amounts of video data can be transmitted from the computer to the screen. The port is trapezoid shaped.

2 _____

This port is for inserting a device which is used to store and transport files from your computer. The port has a small rectangular shape.

3 _____

This is an essential port to use when you want to power your computer with electricity. On a laptop, it can be round or rectangular in shape and is used when you want to charge the battery.

4 _____

This port is small and round and is used to output sound from the computer. You can attach a speaker or headphones to your computer using this port.

5 _____

This port is also small and round, but it is used to receive audio signals via a microphone.

6 _____

This port is used to connect a cable from the computer to another screen, router or printer. The port has small screws and either holes or pins, and it can be blue. It is rectangular in shape.

7 _____

This port allows the user to connect the computer to an Internet router. It is a very small port.

Questions

1 Which port is responsible for controlling the sounds released by the computer?

2 Which port comes in two different shapes?

3 Which of the seven ports are available in your computer or a computer you have used?

4 Order the seven ports from the most important to the least important to you.



8 Robots

A Dialogue

1 Read the dialogue and answer the questions.

Teacher: Today we are going to learn about robots. Can anyone name a type of robot that you know about?

Fuad: Yes sir, androids are a type of robot.

Teacher: That's correct, Fuad! Androids are humanoid robots. In other words, they are robots that look and act like humans. What other types of robots can you think of?

Ahmed: Excuse me, sir. I thought that robots were only androids. What exactly is a robot? What does it do?

Teacher: Well, Ahmed, a lot of people think like you, but actually, robots are everywhere. In general, a robot is a machine that can carry out difficult or routine tasks automatically. It can do the job more quickly than a human can through its software, and this is why robots have become so popular.

Ahmed: Mum has an automatic vacuum cleaner. Do you consider this a robot?

Teacher: Of course! That's a robotic vacuum cleaner.

Fuad: I've also heard of industrial robots.

Teacher: That's correct! These machines are found in factories to help with manufacturing goods like cars and electronics.



Questions

- 1 What is a robot?
- 2 What are the types of robot suggested by Fuad and Ahmed?
- 3 Why are robots so popular?
- 4 What other types of robot can you think of?

B Listening

2 Listen and decide whether the following sentences are true or false. Correct the false sentences.

- 1 Robots were first thought of during the Industrial Revolution.
- 2 Leonardo da Vinci made a humanoid robot in 1495.
- 3 'Eric' was the first humanoid robot exhibited in a museum in London.
- 4 Robot technology continues to advance slowly.
- 5 Robots are not very widely used these days.

C Vocabulary

3 Match the following terms related to robots with their meanings.

- | | |
|--------------|--|
| a humanoid | 1 related to manufacturing or building |
| b industrial | 2 quickly |
| c machine | 3 looking like a human in shape |
| d rapidly | 4 an appliance that is built to carry out a task |
| e advance | 5 to move forward and become better |

4 Choose the correct word in the following sentences.

- 1 (An android / A robot) is an automated machine that looks like a human.
- 2 What makes robots so (modern / popular) is their ability to finish work very quickly.
- 3 A machine can complete a (software / task) much more quickly than a human can.
- 4 Machines are built to function (rapidly / automatically) without the assistance of people.

D Grammar

Third Conditional

The third conditional is a structure used when talking about unreal situations in the past.

Like the other conditionals, a third conditional sentence consists of two clauses: an “if” clause and a main clause:

if clause	main clause	explanation
<i>If I had studied harder,</i>	<i>I would have passed the exam.</i>	<i>I failed the exam because I didn't study hard enough.</i>

If the “if” clause comes first, a comma is usually used. If the “if” clause comes second, there is no need for a comma.

Example

*If we had made use of robots in our factory, the production would have been much faster.
I wouldn't have broken the machine if I had fused it carefully.*

Form of third conditional:

if clause = *if* + subject + past perfect verb

main clause = subject + *would* (OR *could* OR *might*) *have* + past participle

5 Join these sentences using *if*.

- 1 He bought a robotic vacuum cleaner. He kept his house clean all the time.
- 2 We couldn't mend the machine. We didn't have any equipment.
- 3 Scientists didn't create robotic bees. They didn't know anything about the planet Mars.
- 4 Car manufacturers didn't use robots in their factories. Car production was very slow and difficult.
- 5 The hospital had robots for sorting medicines. The process was more efficient and much faster.

6 Make third conditional sentences using the phrases below.

- 1 We wouldn't have been able to build a robot _____
- 2 If students hadn't cleaned their school well, _____
- 3 If we hadn't worked all day long, _____
- 4 We would have used an android _____
- 5 If we had used robots in factories, _____

E Reading

7 Read the text and answer the questions.

Robots: Past and Present

The first ever robot known to be built was created around 400–350 BCE by Archytas. Archytas was a very clever man who was an excellent mathematician, philosopher and astronomer from Ancient Greece. His robot was a mechanical bird made out of wood and powered by steam. The body of the bird was a hollow cylindrical shape that had wings on each side and smaller wings towards the back. The front was pointed like a bird's beak. Near the back of the bird, there was an opening which was connected to a boiler that made steam. The pressure of the steam allowed the bird to take flight. The robot bird was able to flap its wings and fly up to 200 metres. It is believed that the bird was attached to a cable and that a pulley and counterweight helped it to fly.

Robots today are much more sophisticated and complex, although scientists are still using the shapes of animals in their designs. The 'MantaDroid' is a robot which is shaped like a manta ray and can swim through the sea and gather data about marine life and do underwater searches. The 'Snakebot' is a robot shaped like a snake which can search sewers and earthquake sites. There is even an 'Octobot' which is soft like a real octopus as it is made of silicon. Its tentacles are powered by chemical reactions in 3D-printed chambers under its skin.

Questions

- 1 When was the first known robot made?
- 2 What was the first known robot made of? How was it powered?
- 3 What is the difference between the first robot and today's robots?
- 4 What is the 'MantaDroid' used for?
- 5 How are the tentacles of the 'Octobot' powered?



F Writing

- 8 Imagine that robots had never been discovered or developed. Write a paragraph describing how life was in the past before the invention of robots and how robots have changed it. Use the following table to organise your ideas.

Title:
Topic Sentence: <i>If robots had never been invented ...</i>
Life in the past without robots ...
Life now with robots ...
Concluding sentence:

Culture and Values 2

Women in Business

Women in Business (WiB) is a nonprofit organisation whose main aim is to **facilitate** trade transactions between the Arab World and UK. This organisation is managed by an international team of unpaid **volunteers** who work hard to realise its ambitions and aspirations.

In addition, WiB is also considered to be a worldwide **forum** in which women involved in business share ideas and broaden their experiences with others in conferences and meetings held in a variety of countries.

Furthermore, WiB has played a major role in encouraging a worldwide interaction between **qualified** employees and business women, with the aim of enhancing women's economic development.

Finally, WiB encourages all its participants, regardless of their professional level, to take part in its **seminars**. It also provides its members with the opportunity to develop by participating in its annual events. When questioned, over four-fifths of the participants stated that they had benefited from participating in WiB's seminars which were conducted by proficient lecturers and included real-life experiences.

Questions

- 1 What is the main idea of the text?
- 2 What is the meaning of the words in bold?
- 3 Do you think WiB can achieve all its goals?
- 4 In your own words, fill in the following company profile for WiB.

Profile of _____

(Write a brief paragraph that describes the company, its history, the sector in which it operates, etc.)

Company Details

Location: _____

Financial facts: _____

How does the company compare to others in its sector? _____

Personnel: _____

9 X-rays

A Dialogue

- 1 Read the dialogue and answer the questions.

Salma is having an X-ray of her arm taken by a radiographer.

Radiographer: Please place your arm here on the machine, and keep it perfectly still so that we can take the X-ray successfully!

Salma: Okay. You won't be standing next to me, will you?

Radiographer: No, I won't, but don't worry, it won't take long.

Salma: Can you tell me how the X-ray machine works?

Radiographer: Yes, of course. The short wavelength of the X-rays can pass through objects. The machine will produce a very strong X-ray beam, which will travel through your arm. An image will then be made on a metal film.

Salma: The X-ray only shows my bone, doesn't it? I mean, why does it not show the rest of my arm, for example my skin?

Radiographer: This is because X-rays can go through dense things, such as your body tissue and skin. Your bones, however, are hard so the X-rays can't pass through. The soft tissue shows up as black on the X-ray, and your bones show up as white, as your bones absorb the X-rays ... Great! No broken bones!

Questions

- 1 In your own words, describe how the machine takes an X-ray of Salma's arm.
- 2 Why does the X-ray beam pass through the soft tissue of the arm but not the bone?
- 3 What does the soft tissue look like on an X-ray?



B Listening

- 2  Listen and complete the text.

The History of the X-ray

German (1) _____ Wilhelm Conrad Röntgen discovered X-rays in 1895 while he was working in his laboratory. When he applied a high voltage in a (2) _____ with positive and negative electrodes, a (3) _____ green glow appeared. Röntgen covered the tube with heavy black paper and noticed that some crystals that were a distance away from the tube were glowing. He realised that the ray could pass through all solid objects apart from (4) _____. To prove this theory, Röntgen took a film of his wife's hand and also a picture of a set of weights inside a box. Other scientists were very excited by this discovery and tried to (5) _____ the experiment. After only one month of the discovery, X-rays were used in the field of (6) _____ in America and Europe. Today, advances in technology have enabled the production of small, light X-ray machines that produce high-quality images electronically.

C Vocabulary

- 3 Use a dictionary to define these three types of X-rays. Then, put the words in the box under the correct heading.

radiography solar telescope cancer cells treatment examination Earth's atmosphere
bones tumour cure soft tissues satellite

X-ray imaging	X-ray therapy	X-ray astronomy



D Grammar

Tag Questions

Tag questions are used in speech at the end of sentences, often to check information or opinions. Usually, positive sentences have negative tags. Negative sentences have positive tags.

Examples

You can swim, can't you?

You won't be here tomorrow, will you?

4 Complete these sentences with the right tag question.

- 1 Taking an X-ray of my arm will tell me if it's broken, _____?
- 2 The doctor says I shouldn't have too many X-rays as they can be harmful, _____?
- 3 The vacuum inside the X-ray machine allows the electrons to move around, _____?
- 4 The patient didn't keep her arm still when she took the X-ray, _____?
- 5 The doctors wouldn't be able to see my skin on the X-ray, _____?
- 6 It's because bones are hard that X-rays can't pass through, _____?
- 7 The X-rays are shorter than normal light rays, _____?
- 8 They have encountered problems with the X-ray machine before, _____?
- 9 The doctors will complain to the hospital manager about the constant breakdown of the X-ray machine, _____?
- 10 The radiographers did not expect the hospital to receive a new X-ray machine, _____?



E Reading

5 Read the text and answer the questions.

Medical X-Rays

X-rays are widely used in the medical setting. They are able to detect broken bones, as well as other injuries and diseases in the body. X-ray machines are a highly valuable diagnostic tool. Radiography is the process of taking an image called radiograph, using X-rays. The X-ray machine takes a picture on a special film, or electronically in more modern machines. This type of X-ray is commonly used, for example in chest X-rays and dental X-rays.

Fluoroscopy uses X-rays to examine flowing blood and also the intestine and the heart. Patients with suspected stomach problems can drink a liquid called 'barium meal' and moving images of their stomachs can be taken. To detect heart issues, a special dye can be injected into the blood stream so that images can be taken. This procedure is called an angiogram.

Finally, a CT scan is a more advanced X-ray test. The patient lies down inside a big scanner and lots of tiny X-ray beams are directed to the part of the body with the symptom. The scanner moves slowly to allow lots of pictures to be taken from different angles. These images are displayed on a TV screen.

Questions

- 1 What is the author's purpose in writing this text?
- 2 List the three types of X-ray procedures mentioned in the text.
- 3 Put the following quote, taken from the text, into your own words: "X-ray machines are a highly valuable diagnostic tool."
- 4 Think of another type of machine that can detect illness in the body which is not mentioned in the text.
- 5 Find words in the text that mean:
 - a to discover or notice
 - b a problem in the body that indicates a certain condition or disease
 - c to check in detail in order to determine a certain condition or disease
 - d an image of the heart



CT scan

F Writing

- 6 With the introduction of the X-ray, certain aspects of our lives have changed. Write a paragraph about these benefits and add some of your own.

Review (Units 7-9)

A Match the following words with their meanings.

- | | |
|------------------|---|
| 1 discover | a to advise or suggest something |
| 2 recommend | b supplied with mechanical or electrical energy |
| 3 technology | c an expert in technical equipment |
| 4 technician | d a way of doing things |
| 5 sophisticated | e to find something for the first time |
| 6 powered | f new methods, products, etc. |
| 7 patient | g developed to a high degree of complexity |
| 8 procedure | h an ill person |
| 9 innovation | i computers, electronics, machinery, robotics, etc. |
| 10 significantly | j in a notable way |

B Fill in the blanks with a word from the box.

detect medical wavelength absorb
dense vital bones

An X-ray machine is a (1) _____ piece of equipment used widely in the (2) _____ setting. It is used to (3) _____ broken bones and diseases in the body. The X-rays have a short (4) _____ which can pass through (5) _____ objects such as skin and body tissue. However, X-rays cannot pass through hard objects such as (6) _____, which (7) _____ the X-rays. This is why the bones can be seen as white on the X-ray.

C Construct a sentence with each word.

- 1 connect _____
- 2 android _____
- 3 radiograph _____
- 4 examination _____

D Join the following phrases to make a meaningful paragraph.

computer technology, hard drives, faster and smaller, improve efficiency, totally wireless, potential for further advancements, biometrically, perceptual computing

E Complete the clauses.

- 1 Robots today are sophisticated and complex and can be found in _____.
- 2 An android robot has a _____.
- 3 Robots are very useful in manufacturing and industry as they can _____.

F Write true or false. Correct the false ones.

- 1 Computer technology continues to advance at a rapid pace.
- 2 We do not see computers very often on a day-to-day basis.
- 3 Computer manufacturers are not responsible for protecting the environment and can cause a build-up of dangerous materials.
- 4 A computer with a large RAM would be more suitable if you wanted to play computer games.

G Fill in the blanks with the correct form of the verbs in brackets.

- 1 If Malik didn't know about computers, he (work, not) _____ in the computer shop.
- 2 If the student's father was familiar with computers, he (choose) _____ one by himself.
- 3 If Salah didn't prefer working for our company, he (be, still) _____ working in carpentry.
- 4 If Fatima's job (have) _____ more disadvantages than advantages, she would quit it.
- 5 If I had a laptop, I (love) _____ to do all my school projects electronically.
- 6 You (use) _____ a video card to speed up your computer if you had one.
- 7 Karim (upload) _____ his photos onto his laptop if he had known how.
- 8 If my computer (be) _____ broken, I would ask a computer expert to fix it.
- 9 I (consult) _____ the technician in the shop where I bought my computer if I had had any questions.
- 10 If I were a specialist in computer manufacturing, I (design) _____ the smallest, lightest and cheapest computer.

H Join these sentences using *if*.

- 1 I didn't buy a computer last year. I didn't exchange it for a laptop.
- 2 Alia was a nurse. She liked being a doctor more.
- 3 I learned how to use the X-ray machine. I took radiographs of people's body parts.
- 4 The X-ray machine broke down. We called the head radiographer immediately.
- 5 Industries needed to work more quickly. They invented robots.
- 6 Robots minimised the amount of repetitive jobs humans need to do. People started to do more interesting jobs.
- 7 Industries were concerned about the danger of working in certain environments. They created robots to do the jobs instead.

I Complete these sentences with the right tag question.

- 1 We have to choose a computer company to install the office computers, _____?
- 2 I'm very lucky to own this robotic vacuum cleaner, _____?
- 3 You prefer working on a laptop, _____?
- 4 This robot has moveable parts, _____?
- 5 An X-ray machine is useful for detecting broken bones, _____?
- 6 Having an X-ray doesn't take much time, _____?
- 7 Laptops should have a protective case, _____?

10 Automatic Control

A Dialogue

- 1 Read the dialogue between Tarek and a door-to-door salesman selling a manual shower head then answer the questions.

Salesman: Can I interest you in a shower head?

Tarek: I already have a shower head in working condition.

Salesman: But this shower head is unlike any other. It runs by a manual control system and so minimises any possibility of something going wrong.

Tarek: So, what you're telling me is that I have to manually work the shower head. That doesn't sound as easy as my automatically-controlled shower head. With the press of a button, it is able to change from running hot water to cold.

Salesman: Automatic control systems are much more comfortable, but this manual shower head can save you a lot on repair expenses!

Questions

- 1 What is the salesman trying to sell Tarek?
- 2 Why wasn't Tarek interested in buying this item?
- 3 Decide if the sentences are true (T) or false (F). Correct the false sentences.
 - a A manual shower head runs hot or cold water at the press of a button.
 - b Comfort and ease come with automatic control systems.
 - c The benefit of manual control systems is that they reduce the possibility of sudden malfunctions.
 - d Maintenance of automatic systems is cheaper than the maintenance of manual systems.

B Listening

- 2  Listen to the text about automatic control systems and complete the sentences.

- 1 In a society greatly based on industry, control systems _____.
- 2 Control systems deal with _____.
- 3 Some examples of these control systems include the _____.

C Vocabulary

- 3 Complete the sentences by using words from the box. Use the definitions beneath each sentence as a clue.

control systems automatically response
processes manual

- The doors opened _____ as we approached.
(a device working without direct human control)
- It would take too long to do a _____ search of all the information.
(operated or done by hand or without the help of electricity, computers, etc.)
- Most mechanisms today are run by _____ that manage their functions.
(the organisation of parts to control a function)
- Industries carry out many _____ to produce their materials.
(a series of actions taken in order to reach a particular result)
- The _____ of the automatic control system of the toaster was to eject the bread once it had finished toasting it.
(something that is done as a reaction to something that has already happened)

D Grammar

Indefinite Pronouns
An indefinite pronoun does not refer to any specific person, thing or amount. It is vague and "not definite". Most indefinite pronouns are either singular or plural. However, some of them can be singular in one context and plural in another.
Singular <i>another, anybody / anyone, anything, each, either, enough, everybody / everyone, everything, less, little, much, neither, nobody / no one, nothing, nowhere, one, other, somebody / someone, something, somewhere</i>
Plural <i>both, few, many, others, several</i>
Singular or plural <i>all, any, more, most, none, some, such</i>

- 4 Fill in each blank with a suitable indefinite pronoun from the table above. Some sentences may have more than one correct answer.

- I know _____ about how to solve telecom problems, but you must promise to keep it a secret.
- _____ knows him. He is a stranger who claims to be the new employee at the Telecom department.
- I had dinner _____ near the Telecom Service building.
- _____ could have known that telecom is a great tool in the facilitation of business growth.
- _____ can exceed the capacity of telecom to connect businesses and investors.
- This company is _____ near considering hiring new telecom employees. Let's go _____ else and look for more job opportunities.
- Would you like _____ to wash your hands with?
- This is _____ all telecom students must learn.
- They took him _____ in London as a manager in telecom services.

E Speaking

5 Work in pairs. Discuss with your partner the difficulties you would face if automatic control systems were never invented. Start off with the statements below.

- *If automatic control systems had never been invented, I feel I would have a hard time ... (doing this)*
- *I agree / disagree because ...*

F Reading

6 Read the text and the fact box, then answer the questions on page 53.

The Development of Automatic Control Systems

Automatic control has greatly influenced the development of automation or machines. It originated in the studies and inventions of the ancient world. People used to organise a functional system based on water and level work to make jobs, like pulling up a wooden gate, easier.

From the 17th century onwards, such systems developed into ones designed to control the temperatures and the mechanics of mills and steam engines. Similarly in the 19th century, control systems were modified to stabilise and steer ships and planes.

With the introduction of communications engineering in the 1930s, control systems began to develop into the automatic processors we know today.

FACT BOX

- Control systems became automatic
- Basic functional systems led to the invention of control systems
- Control systems used to guide large vehicles
- Basic functional systems were updated to control mills and steam engines

Automatic Control

Questions

1 Fill in the timeline with facts from the box. Refer to the text for help.

Ancient world: _____.

17th century: _____.

19th century: _____.

1930s and on: _____.

2 Circle the answer that best completes the sentence.

a Control systems originated as far back as _____.

i the 17th century ii before time iii the ancient world

b _____ control systems can be said to have started around the 1930s.

i Manual ii Automatic iii Functional

c In ancient times, functional systems were used to carry out _____.

i manual labour ii level work iii easy jobs

G Writing

7 Imagine the world today without automatic control systems: people would always have to use the stairs instead of lifts; our windows wouldn't close with a push of a button; a lot of our daily activities would become much more difficult. Write a short paragraph discussing what difficulties you might face without the invention of automatic control systems.



manual shower head

11 Electrical Power Stations

A Dialogue

- 1 Read the dialogue and answer the questions.

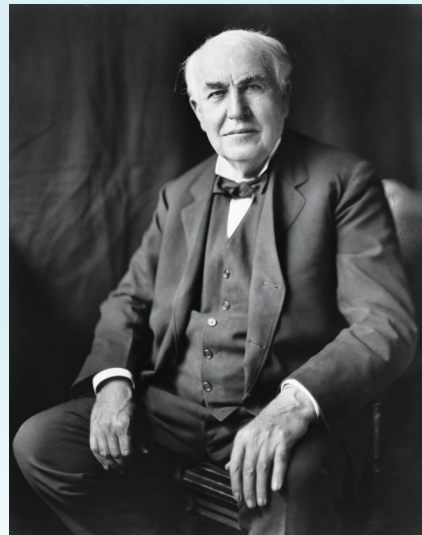
Hassan is a young man from the present day who has gone back in time to interview Thomas Edison (the inventor who built the first power stations).

- Hassan:** I'm honoured to meet you Mr Edison. Your discovery has changed the world!
- T. Edison:** I'm glad you're here because I have some questions of my own. How has the first power station, which I built in 1882, developed?
- Hassan:** Well, electrical power stations have become vital all over the world. Improvements have also been made such as the types of sources of energy that are being used.
- T. Edison:** What do you mean by sources of energy? Have power plants been able to find alternative forms of fuel to generate electricity?
- Hassan:** Why yes, they have! Not only do they use gas and oil but also coal and fossil fuels. We even have different power stations nowadays.
- T. Edison:** Are they also used to generate electricity?

- Hassan:** Yes, of course! They produce electricity through heat, water, wind and even the sun's heat.
- T. Edison:** I must say, power stations have come a long way.

Questions

- 1 Who is Thomas Edison and why is he so interested in power stations?
- 2 Identify three ways that electrical power stations have changed since 1882.
- 3 How did the changes in electrical power stations influence your life?

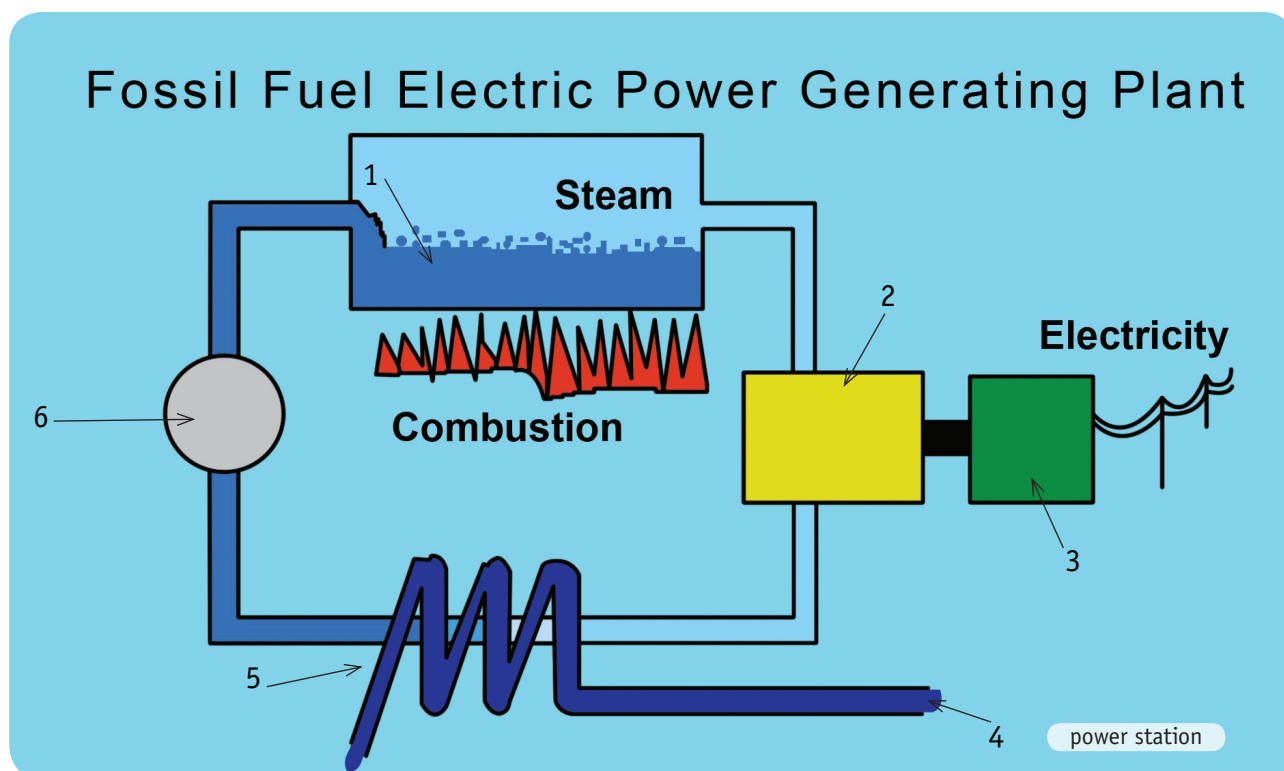


Electrical Power Stations

B Listening

- 2 Listen and label the figure using words from the box.

pump turbine water generator
cooling water condenser



C Vocabulary

- 3 Use a dictionary or thesaurus to look up synonyms for the underlined words in order to complete the sentences which describe the job of a power station operator.

A power station operator holds the position of a facilitator or (1) _____. An operator makes sure his crew or (2) _____ is safe and working. Operators also have to maintain or (3) _____ the equipment by constantly inspecting or (4) _____ important information. They may also control the generators or (5) _____ depending on need. Most importantly, operators should be able to react to an emergency or (6) _____ and carry out the necessary procedures or (7) _____.

D Grammar

Prepositions

Prepositions are used before nouns to give additional information in a sentence. Prepositions are usually used to show where something is located, when something happened or an action.

Prepositions showing:

Location:

above, below, over, under, among, between, beside, in front of, behind, next to, with, in the middle of, on, in, at

Time:

at, on, by, before, from, since, for, during, to, until, after

Action and Movement:

at, by, from, into, on, onto, off, out of, towards

4 Choose the correct answer for each question.

- 1 He made his escape by jumping _____ a window and jumping _____ a waiting car.
a over / into b between / into c out of / between d out of / into
- 2 To get to the Operations Department, you have to go _____ those stairs and then _____ the corridor to the end.
a up / into b between / into c out of / between d out of / into
- 3 I saw something about it _____ television.
a in b on c at d with
- 4 I couldn't get in _____ the door of the lift so I had to climb _____ the window of the shaft.
a through / out of b between / into c out of / between d out of / into
- 5 She took the key _____ her pocket and put it _____ the lock.
a over / into b between / into c out of / in d by / on
- 6 He drove _____ me without stopping and drove off _____ the centre of town.
a from / into b towards / over c along / up d past / towards
- 7 I took the old card _____ the computer and put _____ the new one.
a through / in b out of / in c out of / between d out of / into
- 8 I went _____ him and asked him about the best way to get _____ town.
a from / into b towards / over c along / up d up to / out of
- 9 Cables run _____ the ground, and water pipes wrap _____.
a through / in b out of / in c under / around d out of / into
- 10 I work _____ a power station _____ the field.
a through / in b out of / in c under / around d in / next to

Electrical Power Stations

E Reading

5 Read the text and answer the questions.

Power stations were first thought of by Thomas Edison and his partner Edward Johnson. These two scientists organised and built the Edison Electric Light Station in January 1882. This station was powered by a steam engine. It provided electricity for the area around, that could be reached through the water pipes without digging up the road. Customers of this power station included the Telegraph Office and the General Post Office.

Today, power stations are set up all over the world. Electricity has become vital in carrying out daily activities. It is supplied through underground and overhead cables that reach any desired distance and location. Power stations have developed so much since Edison and Johnson's time that many now use other forms of energy to generate electricity.

Questions

1 Define the following words from the text:

- a partner
- b steam
- c supply
- d overhead
- e develop
- f form

2 Who thought of the first power station and when?

3 Describe how the first power station was like.

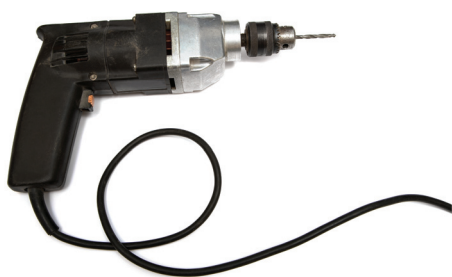
4 Put the following sentence in your own words: "Electricity has become a great necessity to carry out daily activities."

5 Find the antonyms of these words from the text:

- a last
- b destroyed
- c took away
- d consume / use up

F Writing

6 Think about how important electricity is in your daily life. What activities require the use of electricity? How different would your life be if electricity had never been discovered? Write a paragraph discussing how electricity has become a large part of our daily lives. You may refer to facts stated in this lesson or come up with your own.



12 Latest Discoveries

A Dialogue

1 Read and answer the questions.

1

A: Hello. Can I ask you a question, please?

B: Yes.

A: According to you, what was the most important discovery in history?

B: I'd say medicine.

A: Can you be more specific?

B: Vaccines were an important discovery. Because of vaccines, children don't get awful diseases anymore.

A: Thank you for your time.

B: You're welcome.

2

A: What do you think was the most important discovery in history?

C: Well, there have been so many things, but I'd say electricity.

A: Electricity?

C: Yes, I think so. We need it every second of the day, don't we? We need it to use our computers, to browse the Internet, to switch on the lights, for heating and watching television.

A: Right, thanks very much.

3

A: What do you think was the most important discovery in history?

D: Radio waves! We need them for our mobile phones and satellites. I mean, we can't live without them!

A: That's right. Thanks a lot.

D: No problem.

Questions

- 1 What is the interviewer (A) interested in knowing from the three interviewees (B, C and D)?
- 2 What were the three answers to his main question?
- 3 What is the meaning of 'awful'?
- 4 According to you, which one of the three discoveries is the most important? Why?
- 5 How would you answer the interviewer's question? Explain your answer.



B Listening

2 Listen and circle the correct answer.

- 1 Colin is _____.
a sixteen b seventeen c eighteen
- 2 Colin is from _____.
a England b Ireland c Scotland
- 3 Colin's invention is for children with _____.
a breathing problems b hearing problems c learning problems
- 4 The child breathes the spray through the elephant's _____.
a mouth b tail c trunk
- 5 Who suffers from asthma?
a Colin's mother b Colin's cousin c Colin's brother

C Vocabulary

3 Write the words between brackets in the right form.

The first (mechanic) (1) _____ calculator was invented in 1623 and the first (commerce) (2) _____ successful device was produced in 1820.

Many toy inventors have university degrees in engineering, such as (produce) (3) _____ design, mechanical engineering or electronics.

In 1951, the first videotape recorder captured live images from television cameras by (convert) (4) _____ the information into (electricity) (5) _____ impulses and saving the information on magnetic tapes.

Scientists are working on how to (integration) (6) _____ solar cells into mobile phones, allowing them to be powered (continue) (7) _____ on just two hours of sunlight a day.

D Matching

4 Match the following titles with their specific descriptions, then group the inventions into different categories and give them appropriate titles.

- | | |
|-------------------------------------|--|
| a Extraction-Assist Robot | 1 allows the user to lift 200 pounds hundreds of times without getting tired |
| b Tread-Walk | 2 helps carers who work in hospitals and health care institutions. It makes use of the company's robot hand technology, featuring 16 fingers |
| c Perpetual Motion Device | 3 represents a combination of GPS and walking stick and includes 4 mini cameras and a GPS receiver |
| d Phone with Sound Waves | 4 turns sound waves into electricity |
| e Hair-Washing Machine | 5 generates more energy than it consumes and produces power from perpetual motion |
| f Paper-Based Lithium-Ion Battery | 6 is developed for the elderly and disabled in order to enable them to fulfil their personal mobility needs |
| g Raytheon's XOS 2 Exoskeleton | 7 carries injured soldiers out of battlefield, breaks locked doors and lifts heavy cargo |
| h Visually Impaired Assistant (VIA) | 8 is used to power electronic newspapers and to make smart packaging that would assist marketers |

E Grammar

Sentence Fragments and Run ons
A sentence fragment is an incomplete sentence. Some fragments are incomplete because they lack either a subject or a verb, or both. Example <i>Some of the inventions studied in Professor Saleh's class last semester.</i>
A run-on sentence is a sentence in which two or more independent clauses (that is, complete sentences) are joined without appropriate punctuation or conjunction. Example <i>It is nearly half past five we cannot finish the machine before dark.</i>

- 5 Mark complete sentences with CS, sentence fragments with F and run-on sentences with RO.

- _____ 1 The invention that I saw yesterday.
- _____ 2 It is not an exceptional discovery, it is only average.
- _____ 3 Your idea to invent a time machine is quite impossible.
- _____ 4 Because I have no new inventions yet.
- _____ 5 Present your new invention carefully.
- _____ 6 Our latest discoveries were revolutionary, therefore, our profits will be higher.
- _____ 7 Having finished her new project, Joy sat down to an evening of television.
- _____ 8 The steam engine, one of the well-known inventions in history.
- _____ 9 I went in, I adjusted a few details. The machine worked perfectly well.
- _____ 10 Hurry up!

- 6 Correct the run-on sentences in exercise 5.

- 7 Correct the fragments in exercise 5.

F Speaking

- 8 The following is a list of six new inventions with a short description of each. In groups of four, discuss which one is the most important and which is the least important.

Plastic into Oil

This machine is able to turn plastic into oil; it will help people use more renewable energy sources.

Robot Suit

This is a robot suit that elderly farmers can attach to their bodies to help them carry out their daily work more easily.

Toothbrush without Toothpaste

This is a toothbrush that creates a solar-powered chemical reaction inside a person's mouth and cleans his / her teeth without using the toothpaste.

- 9 Prepare a five-minute talk to explain your answer to task 8.

G Reading

10 Read the text and answer the questions.

Self-Cleaning Clothing

Wool manufacturers are interested in this invention: coating fibres such as wool with titanium dioxide nanocrystals, which break down food stains when exposed to sunlight. Of course, these nanocrystals do not harm the skin. They don't even affect the feel or look of the fabric. When coated with titanium dioxide, fabrics become self-cleaning and may be useful in hospitals, for instance.

Some may say that this is not a new invention since materials that use nanoparticles are already on the market. However, such materials are stain-repellent fabrics with a nanofibre coating that pushes the liquid off the fabric by making the particles roll off instead of being absorbed into the material.

Titanium dioxide can decompose organic matter when exposed to ultraviolet light and when mixed with water. It can also destroy bacteria when exposed to sunlight.

The most important feature of titanium dioxide is that it can get rid of ink stains in seventeen hours and coffee evaporates in two hours.

Questions

- 1 Write a suitable introduction to this text.
- 2 Find five difficult words and explain them using the glossary.
- 3 Give an example of stain-repellent fabrics.
- 4 Explain, in your own words, the difference between the type of fabric described in the text and stain-repellent fabrics.
- 5 Write a suitable conclusion to this text.



H Writing

- 11** Use the Internet to find two recent inventions you are interested in and write a fully-developed paragraph describing each.

Culture and Values 3

Desertification

Desertification is one of the most dangerous environmental problems facing our world today. Around a sixth of the world's population has to face its consequences. Countries with a dry climate offer the perfect environment for desertification. In such areas, the surface of the soil is ruined and so no plants grow. Consequently, no animals can graze there. These harsh conditions force people to leave their land and go to other areas where they can get water and food. Some people survive their journey, while others die of starvation or thirst.

Desertification is not only caused by natural conditions. People also play a vital role in accelerating the process of desertification. Farmers tend to overgraze the land in order to feed the growing population. Moreover, people are cutting down trees and this erodes the soil since there are no more trees to protect it against the wind and rain.

The effects of desertification are not as simple as they might seem. Strong winds and wild fires often occur in desert areas, and this might have a serious impact on the availability of the most valuable resource on the planet: water.

Questions

- 1 What is the problem described in the article?
- 2 What are the causes of the problem?
- 3 Choose the correct endings to the following sentence beginnings.
 - a Land can no longer be used for growing crops if ...
 - 1 the top layer of soil is destroyed.
 - 2 animals are grazing on the land.
 - 3 people move to greener areas.
 - b The Earth's most precious resource is ...
 - 1 productive land.
 - 2 water.
 - 3 people.
 - c Trees and plants ...
 - 1 grow well in deserts.
 - 2 stop wind and rain from damaging land.
 - 3 are a major cause of desertification.
- 4 Research another environmental problem and compare it to desertification in a Venn diagram.

Review (Units 10-12)

A Fill in the blank with a word from the box.

processed renewable electricity
gathered burned

Biomass production is another way of producing **(1)** _____ by using rubbish or other **(2)** _____ reserves such as corn or other vegetation. Methane produced by garbage decomposition is **(3)** _____ in ducts and later **(4)** _____ to produce electricity. The burning of vegetation and wood instantaneously produces energy like fossil fuels; otherwise, they can be **(5)** _____ to form beverages.

B Match the words with their synonyms.

- | | |
|-------------------|------------------|
| 1 instantaneously | a make false use |
| 2 exploit | b eject |
| 3 discharge | c apply |
| 4 exert | d immediately |

C Join the following phrases to construct a meaningful paragraph.

manual control system, minimise, going wrong, press of a button, automatic control system, comfortable, save, repair expenses

D Use each word in a meaningful sentence.

- 1 vaccine _____
- 2 awful _____
- 3 browse _____
- 4 waves _____
- 5 mobile _____

E Fill in the blanks with suitable indefinite pronouns.

- 1 Manually controlled shower heads are not like any _____ shower heads.
- 2 _____ automatically controlled shower head is more comfortable.
- 3 _____ shower head systems are useful.
- 4 Automatic control has greatly influenced the developments of machines. _____ people have organised a functional system based on primitive ways.
- 5 _____ systems developed to facilitate man's life after the 17th century.
- 6 _____ breakthrough was the introduction of communications engineering in 1930s, that notified the development of automatic processors as we know today.
- 7 _____ would live a comfortable life without the machines endowed with automatic control system.
- 8 _____ can use automatically controlled machines.

F Fill in the blanks with the correct prepositions.

- 1 The mechanic is waiting for you _____ the door.
- 2 The athlete walked _____ the river and back.
- 3 We shall stay in Syria _____ next month.
- 4 The scientists came _____ the laboratory and started working.
- 5 To stay healthy, one should go on a walk _____ the afternoon.
- 6 The driver jumped _____ the river just _____ the accident .
- 7 Laila met the professor _____ the station.
- 8 My family lives _____ a small village.
- 9 Jihad studied _____ Oxford.

G Mark complete sentences with CS, sentence fragments with F and run-on sentences with RO.

- 1 Wool manufacturers interested in self-cleaning clothing.
- 2 Titanium dioxide nanocrystals don't harm the skin.
- 3 When coated with nanocrystals fabrics become self-cleaning.
- 4 Titanium dioxide can destroy bacteria when exposed to sunlight.
- 5 Titanium dioxide can easily ink and coffee stains.

H Correct the run-on sentences in exercise G.

I Correct the fragments in exercise G.

J According to you, what was the greatest discovery in history? Write four sentences about it.
